WORKED SOLUTION

STEP I 1988, Question 9

Pure • Integration by Substitution

STEP I 1988, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s1-q9

The key idea

Both integrals yield to a standard technique once you choose the right piece to work on. In (i) the product of $\ln x$ with a power of x calls for integration by parts, differentiating the logarithm so that it disappears. In (ii) the awkward part is the square root, so we make it the new variable. The $\cos x\,\mathrm{d}x$ is exactly what the substitution needs, and what is left is a rational function that partial fractions can handle.

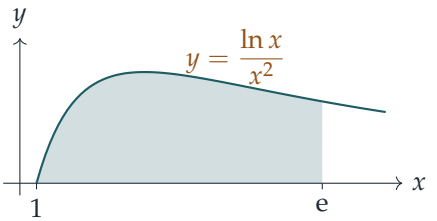
Solution

Part (i)

Integrate by parts with $u = \ln x$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = \dfrac{1}{x^2}$, so that $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{x}$ and $v = -\dfrac{1}{x}$:

$$\begin{aligned}\int_1^{\mathrm{e}} \frac{\ln x}{x^2}\,\mathrm{d}x &= \Big[-\frac{\ln x}{x}\Big]_1^{\mathrm{e}} + \int_1^{\mathrm{e}} \frac{1}{x^2}\,\mathrm{d}x \\ &= \Big(-\frac{1}{\mathrm{e}} - 0\Big) + \Big[-\frac{1}{x}\Big]_1^{\mathrm{e}} \\ &= -\frac{1}{\mathrm{e}} + 1 - \frac{1}{\mathrm{e}} = 1 - \frac{2}{\mathrm{e}}.\end{aligned}$$

We used $\ln \mathrm{e} = 1$ and $\ln 1 = 0$. The answer is positive, as it must be: the integrand is positive on $(1, \mathrm{e}]$.



The shaded area is $1 - 2/\mathrm{e} \approx 0.264$.

Part (ii)

The integrand is defined when $\sin x \neq 0$ and $1 + \sin x > 0$. Put

$$u = \sqrt{1+\sin x}, \qquad \text{so } u^2 = 1+\sin x, \quad 2u\,\mathrm{d}u = \cos x\,\mathrm{d}x, \quad \sin x = u^2 - 1.$$

Here $u > 0$, and $u \neq 1$ because $\sin x \neq 0$. Then

$$\int \frac{\cos x}{\sin x \sqrt{1+\sin x}}\,\mathrm{d}x = \int \frac{2u}{(u^2-1)\,u}\,\mathrm{d}u = \int \frac{2}{u^2-1}\,\mathrm{d}u.$$

> **Ask yourself** What happened to the $\cos x$? Do we have to write it in terms of u?

No. The whole of $\cos x\,\mathrm{d}x$ is replaced by $2u\,\mathrm{d}u$, which is the reason this substitution works so cleanly. Had the numerator been something else, we would have needed $\cos x = \pm\sqrt{1-\sin^2 x}$, with all its sign trouble.

Partial fractions give $\dfrac{2}{u^2-1} = \dfrac{1}{u-1} - \dfrac{1}{u+1}$, so

$$\int \frac{2}{u^2-1}\,\mathrm{d}u = \ln|u-1| - \ln|u+1| + C = \ln\left|\frac{u-1}{u+1}\right| + C.$$

Returning to x,

$$\int \frac{\cos x}{\sin x\sqrt{1+\sin x}}\,\mathrm{d}x = \ln\left|\frac{\sqrt{1+\sin x}-1}{\sqrt{1+\sin x}+1}\right| + C.$$

The modulus signs matter: when $-1 < \sin x < 0$ we have $0 < u < 1$, and $\dfrac{u-1}{u+1}$ is negative. (Strictly, the constant may differ on each interval where the integrand is defined, since the domain is split up by the zeros of $\sin x$.)

An equivalent form comes from multiplying the top and bottom of the fraction by $u-1$. Since $(u+1)(u-1) = u^2 - 1 = \sin x$, the answer is also

$$2\ln|\sqrt{1+\sin x} - 1| - \ln|\sin x| + C.$$

Checks

- Simpson's rule gives $\int_1^{\mathrm{e}} x^{-2}\ln x\,\mathrm{d}x \approx 0.264241$, matching $1 - 2/\mathrm{e} = 0.264241\ldots$.
- For (ii), differentiate the answer: by the chain rule $\dfrac{\mathrm{d}}{\mathrm{d}x}\ln\left|\dfrac{u-1}{u+1}\right| = \dfrac{2}{u^2-1}\cdot\dfrac{\cos x}{2u}$, which is the integrand. Numerically, $\int_{0.3}^{1.2}$ of the integrand is 0.926006 by Simpson's rule, which agrees with the difference of the antiderivative. The agreement is just as good on intervals where $\sin x < 0$, such as $[-1.2, -0.2]$, where the integral is -2.361027.

Common slips

- In (i), choosing $u = 1/x^2$ and $\mathrm{d}v/\mathrm{d}x = \ln x$ instead. This works in principle but is far messier; differentiate the logarithm.
- In (ii), dropping the modulus signs, so the answer is undefined whenever $\sin x < 0$.
- Losing the factor 2 from $2u\,\mathrm{d}u$, which halves the answer.

Beyond STEP

Part (ii) is an instance of a general principle: an integrand that is a rational function of x and $\sqrt{ax+b}$ becomes a rational function of u after the substitution $u = \sqrt{ax+b}$, and rational functions can always be integrated by partial fractions. Here $\sin x$ plays the role of x. In a first-year analysis course you meet the other side of the story: the theorems that justify substitution and integration by parts, and why the constant of integration may differ between the separate intervals of the domain. Bartle and Sherbert, *Introduction to Real Analysis*, covers both carefully.

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